



AWS PARTNER CERTIFICATION READINESS

Program Kick-off & Review

AI Practitioner



Today's Learning Outcomes



During this session, we will cover:

- Program Overview
- About AWS Certifications
- Fundamentals of AI and ML



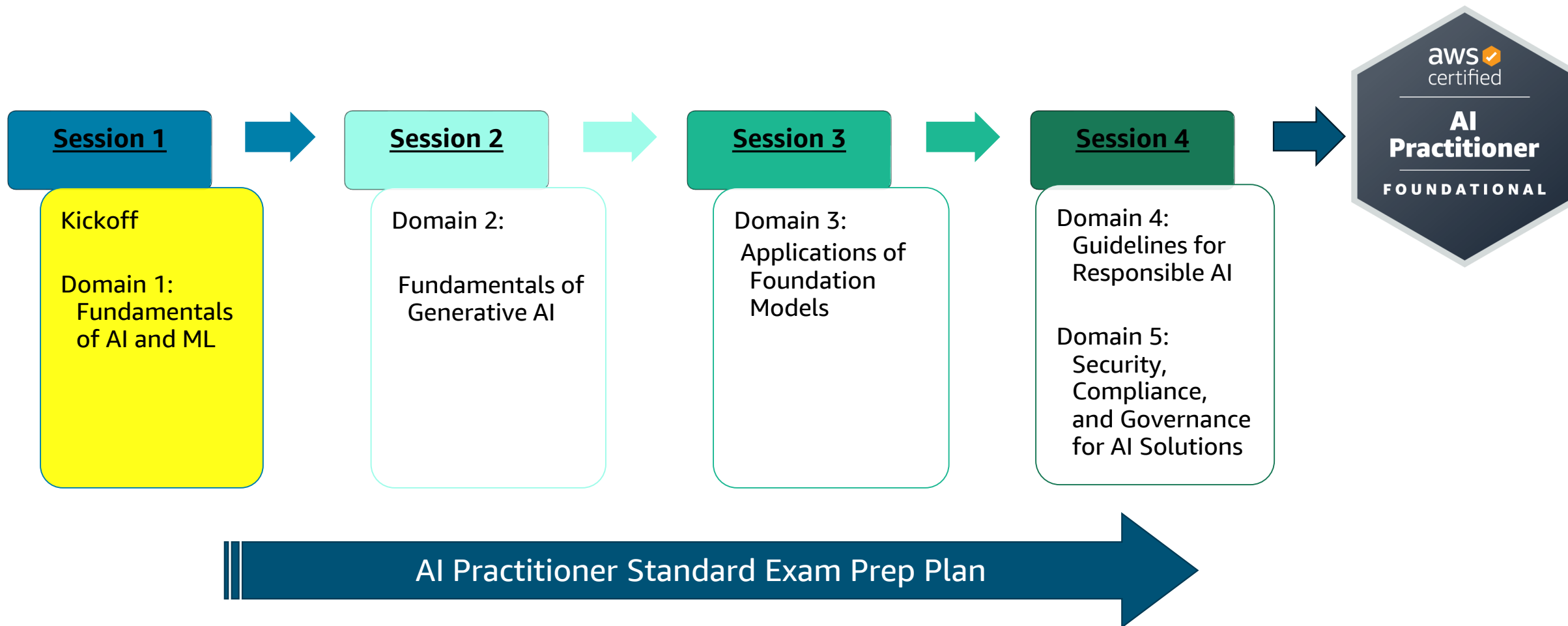


AWS PARTNER CERTIFICATION READINESS

Program Overview



Program Summary



AWS Partner Network (APN) Account

Access AWS Partner exclusive training & link your certification

This program is designed for AWS Partners.

- Register to APN with your company email.
- Link your AWS Certification to this profile.
- Use your APN credentials to access exclusive AWS Partner Training resources



[Click here!](#)

Sign up for APN

First name

Brady

Last name

Smith

Business email

Brady@Company.com

[Verify email](#)

AWS Skill Builder Access

Access digital training and official practice question sets

Utilize for additional training resources.

- Login as **AWS Partner** using APN credentials
- Utilize for supplemental materials
- Utilize for official practice question sets
- Explore and learn!



Click here!

The screenshot shows the AWS Skill Builder login interface. It features four main login options, each with an AWS logo, a title, a description, and a 'Sign in' button. The 'AWS Partner Network' option is highlighted with a red rectangular border. The 'AWS Builder ID' option is marked as 'Recommended for Individuals'. The 'Organization SSO' option includes a text input field for the organization email. The 'Amazon employee single sign-on' option is for users with Amazon work credentials.

Login Method	Recommendation	Sign in Button
AWS Builder ID	Recommended for Individuals	Create or Sign in
Organization SSO		Sign in
AWS Partner Network	For AWS Partners	Sign in
Amazon employee single sign-on		Sign in

OPTIONAL AWS Skill Builder Subscription

The Skill Builder subscription provides access to official AWS Certification practice exams, self-paced digital training content including open-ended challenges, self-paced labs, and game-based learning. **Please note, the Skill Builder subscription is not required for this Accelerator program.**



Free digital training

[LINK HERE](#)

Special features include:

- 600+ digital courses
- Learning plans
- 10 Practice Question Sets
- *AWS Cloud Quest (Foundational)*



Individual subscription

[LINK HERE](#)

Everything in free digital training, plus:

- AWS Cloud Quest (Intermediate - Advanced)
- AWS Certification Official Practice Exams
- Enhanced Exam Prep Courses
- Unlimited access to 1000+ hands-on labs
- AWS Jam Journeys (lab-based challenges)
- AWS Digital Classroom (Annual only)

Individual subscriptions are priced at
\$29 USD per month (*Flexibility to
cancel anytime*) or **\$449 USD per year**.

Access
AI Practitioner
Enhanced Exam
Prep Plan for usage
during this program.

Exam Registration

Set-up your AWS Certification account and register early!

Bookmark the exam page – lots of resources!

- Access CertMetrics through the exam page
- **Recommended** to register with your APN account
- Schedule early and give yourself a deadline

AWS Certified AI Practitioner

Unlock new career possibilities with this AI certification

Schedule your exam

Prepare for the exam



[Click here!](#)



AWS PARTNER CERTIFICATION READINESS

About AWS Certifications



AWS Certifications: identify talent, drive business results

CERTIFIED STAFF DRIVES INCREASED PRODUCTIVITY, SPEED, SECURITY, INNOVATION*

FOUNDATIONAL

Six months of fundamental AWS Cloud and industry knowledge



PROFESSIONAL

Two years of experience designing, operating, and troubleshooting solutions using the AWS Cloud



ASSOCIATE

One year of experience solving problems and implementing solutions using the AWS Cloud



SPECIALTY

Technical AWS Cloud experience in the Specialty domain as specified in the exam guide



*Source: ESG Research Insights Paper Commissioned by AWS, *Understanding the Value of AWS Certifications to Organizations*, October 2020



AWS Certified AI Practitioner

WHAT IT ENTAILS, AND WHO IT IS USEFUL FOR!

AWS Certified AI Practitioner validates in-demand knowledge of artificial intelligence (AI), machine learning (ML), and generative AI concepts and use cases. Sharpen your competitive edge and position yourself for career growth and higher earnings.

Applicable for professionals looking to—

- **Build Skills in AI:** The ideal candidate for this exam is familiar with AI/ML technologies on AWS and uses, but does not necessarily build AI/ML solutions on AWS.
- **Career Advancement:** Advance your career by validating knowledge of AI on AWS





AWS PARTNER CERTIFICATION READINESS

Exam Experience



Taking the exam

- 90 minutes in duration
- 65 Questions (50 Scored questions)
- Scored from 100 to 1,000 (700+ to Pass)
- Multiple Choice, Multiple Response, Ordering, Matching, and Case Study questions
 - No **Negative** points or **Partial** credit
- In-Person and Remote options available
- Live proctor during exam





AWS PARTNER CERTIFICATION READINESS

Exam Contents

AI Practitioner



Domain 1: Fundamentals of AI and ML (20%)

Knowledge of and Skills in:

1. Explain basic AI concepts and terminologies
2. Identify practical use cases for AI
3. Describe the ML development lifecycle



Domain 2: Fundamentals of Generative AI (24%)

Knowledge of and Skills in:

1. Explain the basic concepts of generative AI
2. Understand the capabilities and limitations of generative AI for solving business problems
3. Describe AWS infrastructure and technologies for building generative AI applications



Domain 3: Applications of Foundation Models (28%)

Knowledge of and Skills in:

1. Describe design considerations for applications that use foundation models
2. Choose effective prompt engineering techniques
3. Describe the training and fine-tuning process for foundation models
4. Describe methods to evaluate foundation model performance



Domain 4: Guidelines for Responsible AI (14%)

Knowledge of and Skills in:

1. Explain the development of AI systems that are responsible
2. Recognize the importance of transparent and explainable models



Domain 5: Security, Compliance, and Governance for AI Solutions (14%)

Knowledge of and Skills in:

1. Explain methods to secure AI systems
2. Recognize governance and compliance regulations for AI systems





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Content Review



Section Learning Outcomes



During this session, we will cover:

- Task Statement 1.1: Explain basic AI concepts and terminologies
- Task Statement 1.2: Identify practical use cases for AI
- Task Statement 1.3: Describe the ML development lifecycle



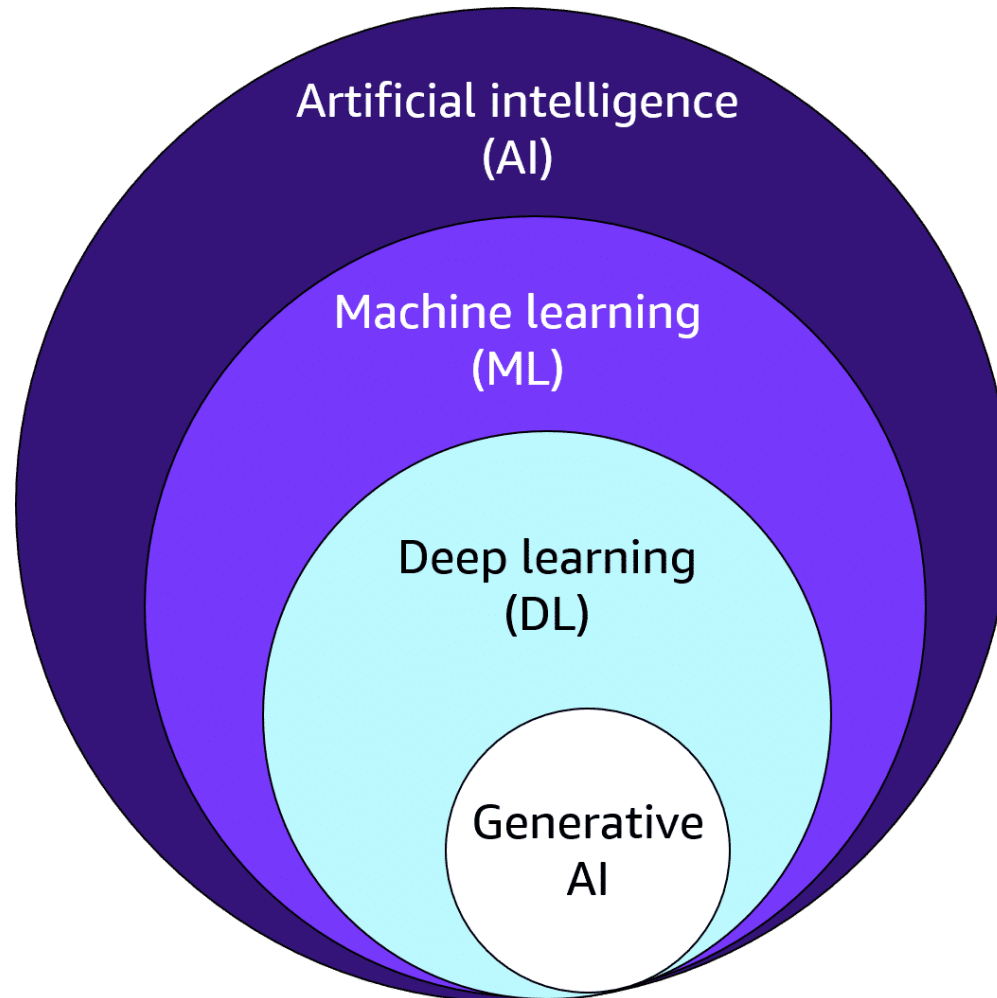
AWS PARTNER CERTIFICATION READINESS

Domain1: Fundamentals of AI and ML

Task Statement 1.1: Explain basic AI concepts and terminologies



Difference between AI, ML, and generative AI

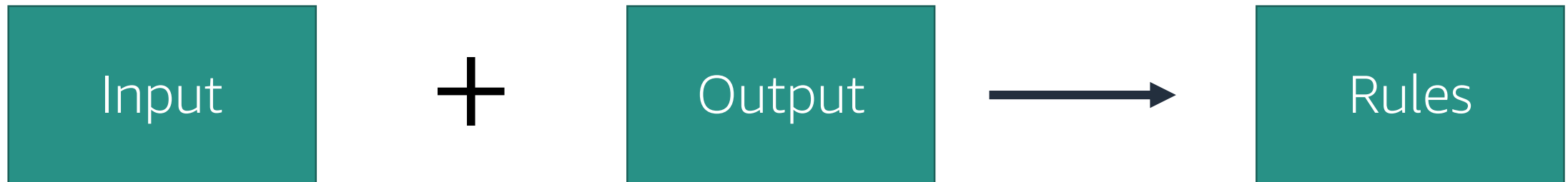


Traditional programming vs machine learning

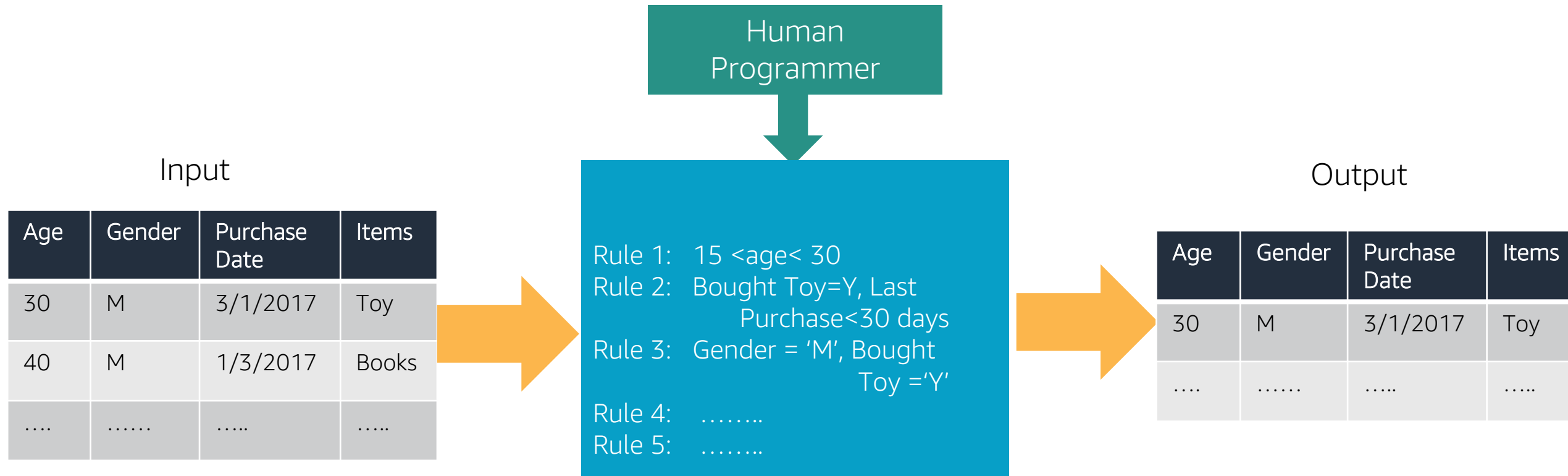
Traditional programming



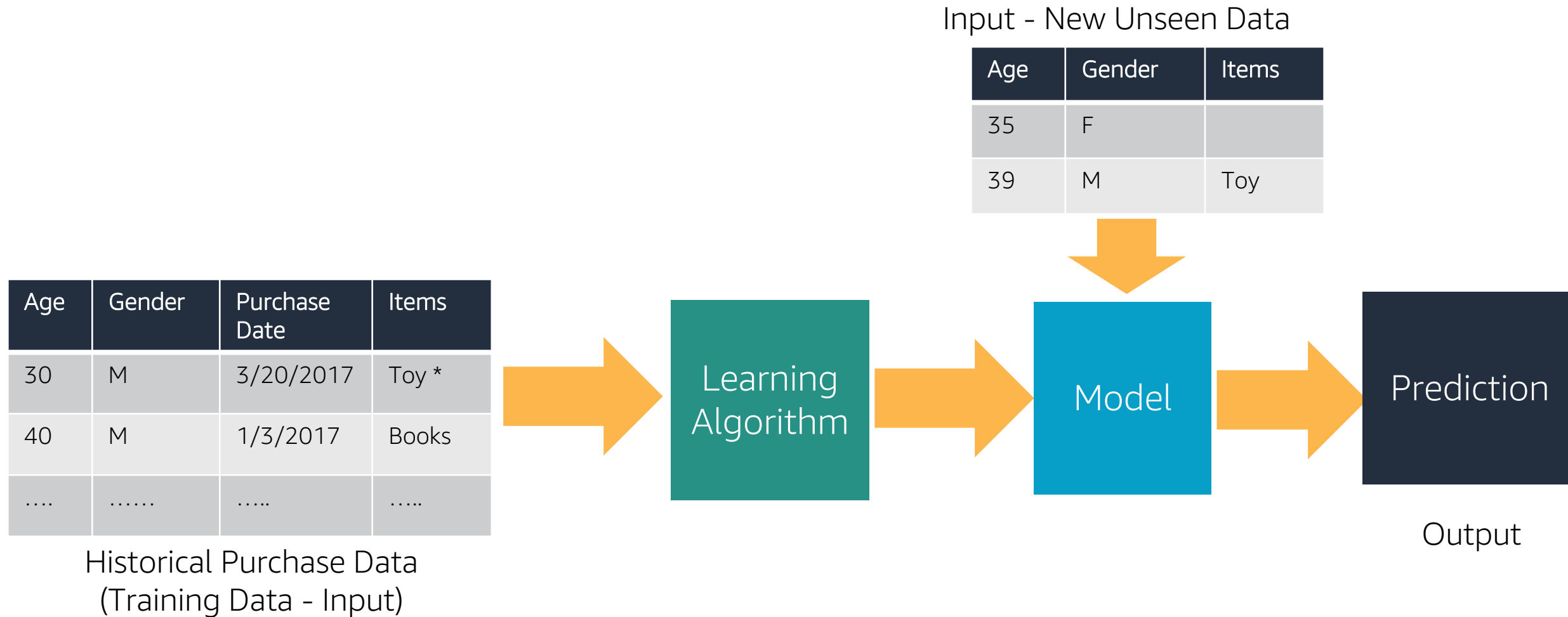
Machine learning



Traditional Programming: You write the business rules



In ML: The data writes the business rules



When to use machine learning

- Use ML when you can't code it
 - Complex tasks where deterministic solutions don't suffice
 - E.g., recognizing speech/images
- Use ML when you can't scale it
 - Replace repetitive tasks needing human-like expertise
 - E.g., recommendations, spam, fraud detection, machine translation
- Use ML when you have to adapt/personalize
 - E.g., recommendation and personalization
- Use ML when you can't track it
 - E.g., automated driving

Why Machine Learning vs Generative AI

Transparency and Interpretability

The decision processes of traditional models are traceable, essential for finance and healthcare sectors.

Explainability and Fairness

Traditional models simplify bias audits in critical sectors like finance, HR, and law through transparent processes

Robustness and Consistency

Traditional models offer consistent outputs, avoiding the nonsensical results sometimes seen in generative AI.

Data Efficiency

Traditional ML models often excel with smaller datasets, unlike large language models needing vast data.

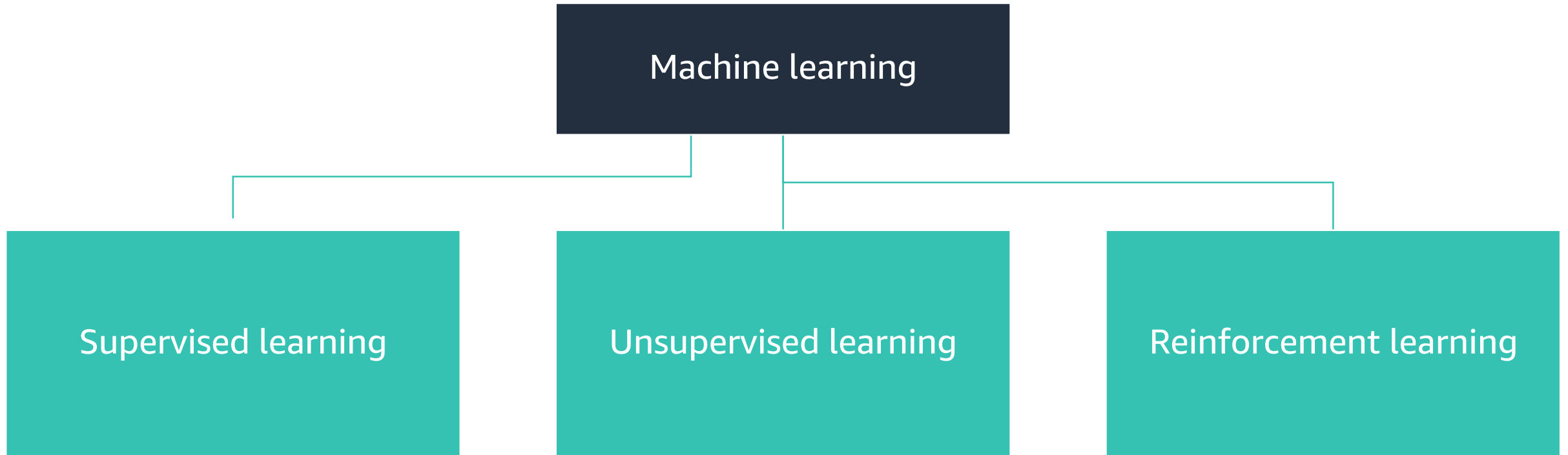
Specific Task Performance

Traditional models, built for specific purposes, can achieve better results in those areas than generative models.

Terminology and concepts

	Statistical Definition	Everyday Definition
Label/Target	Dependent variable	What you are trying to predict
Feature	Independent variable	Data that helps you make predictions
Feature Engineering	Data transformation	Process of reshaping data to get more value out of it
Feature Selection	Variable/subset selection	Process of using the most valuable data

Machine learning types

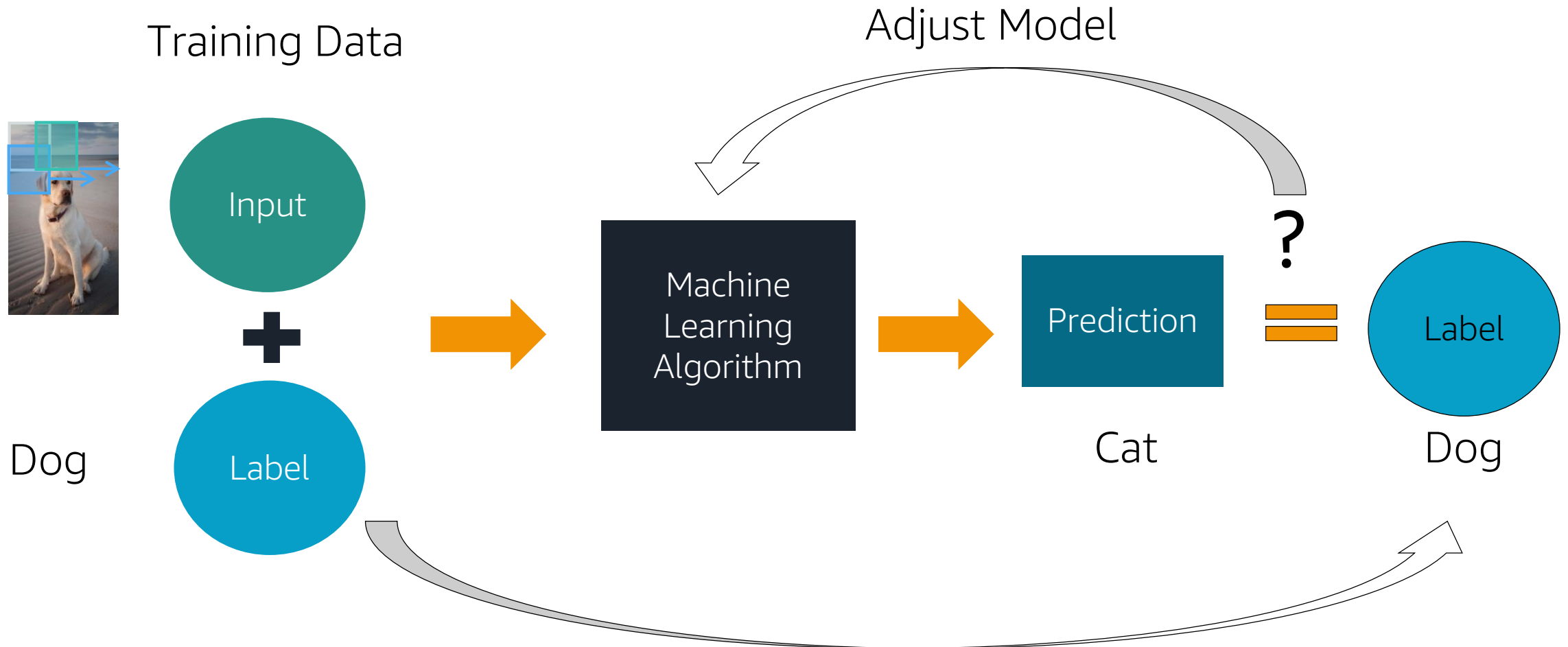


Supervised learning: How humans learn

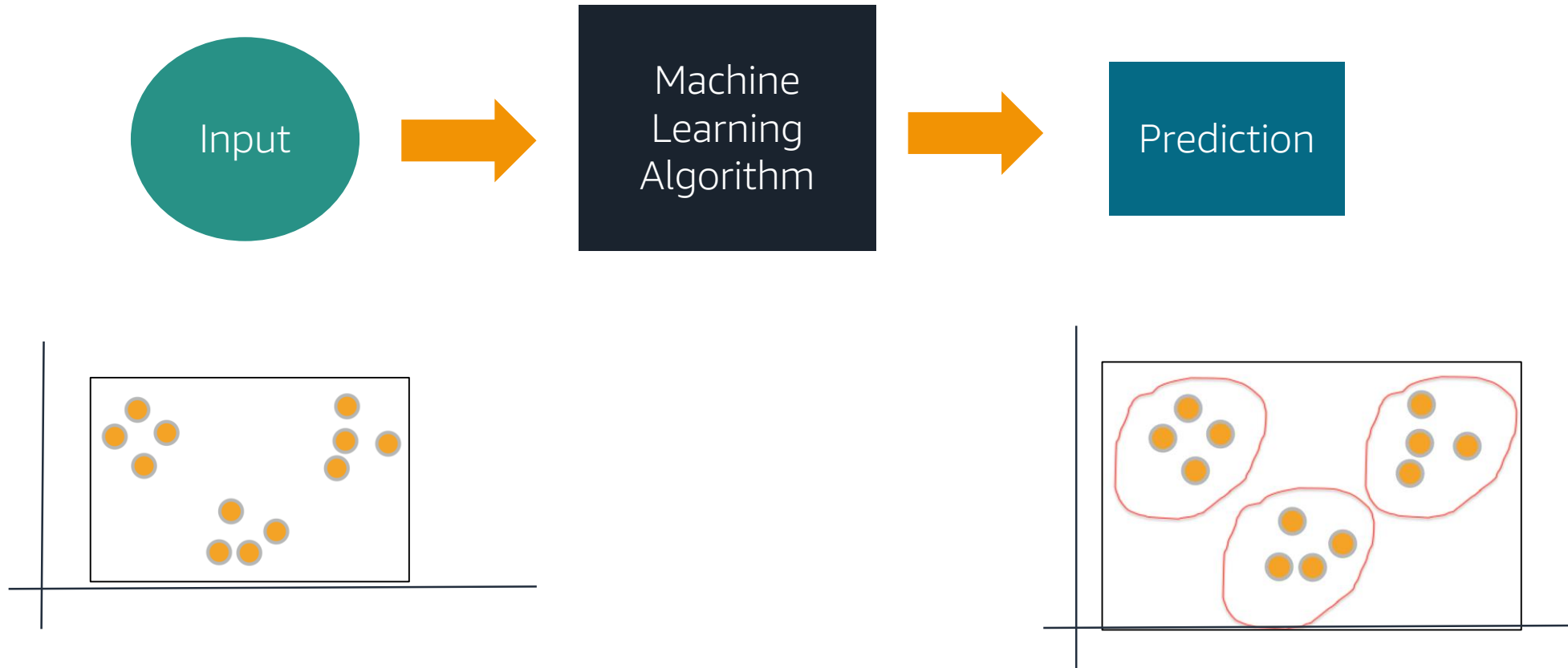


Supervised learning: How machines learn

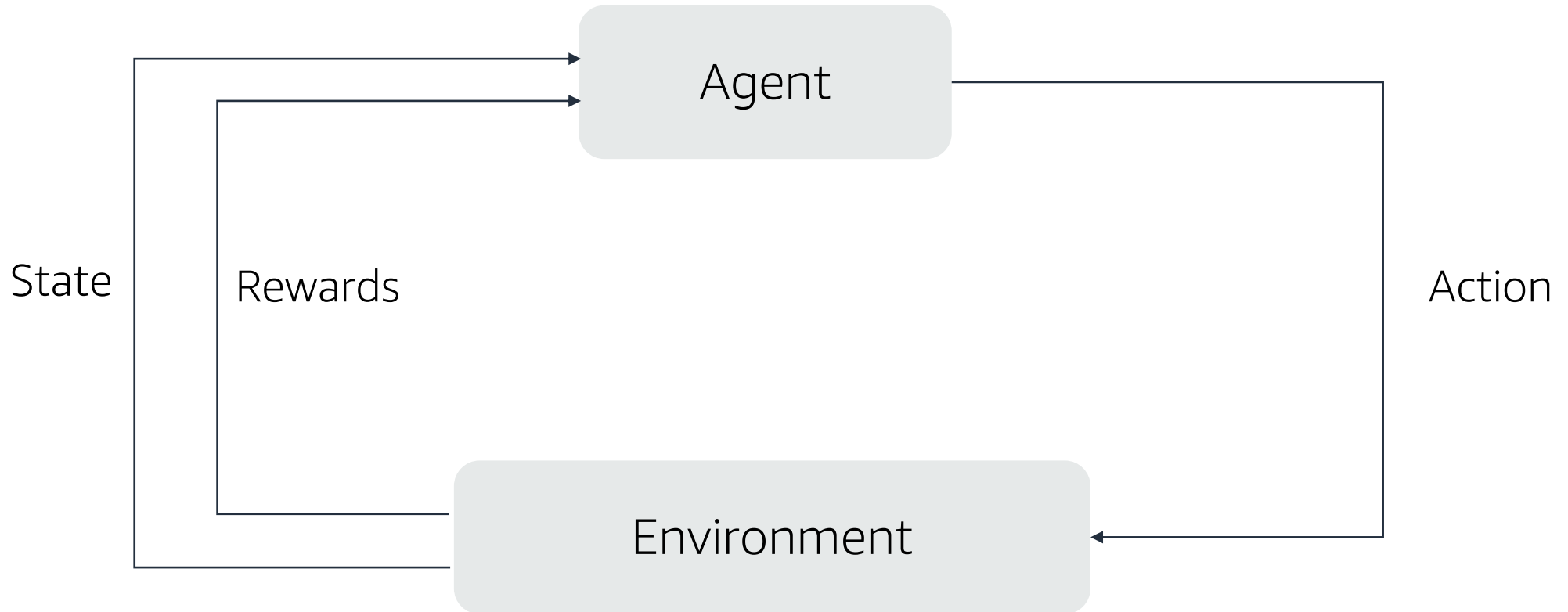
Human intervention and validation required; e.g., photo classification and tagging



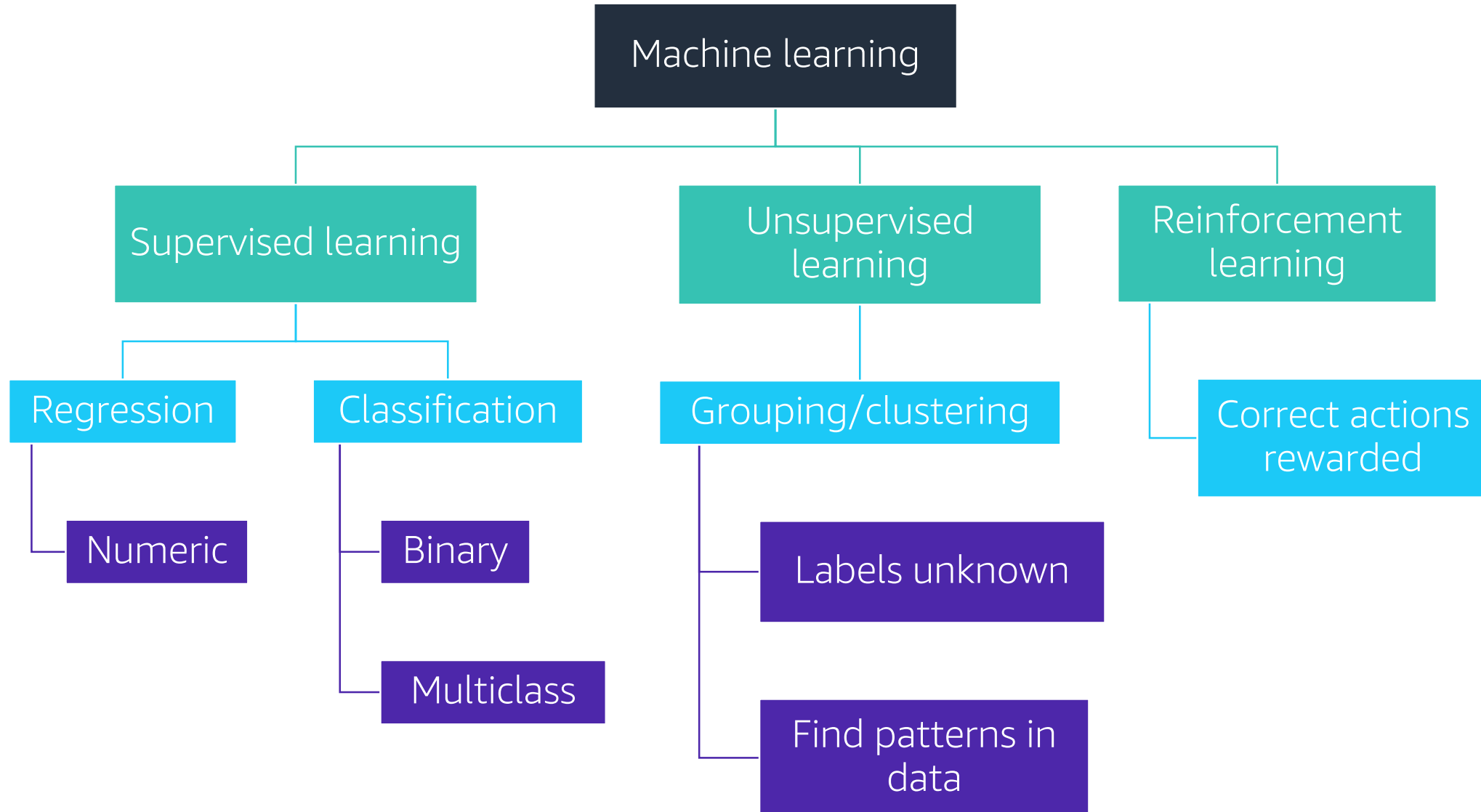
Unsupervised learning



Reinforcement learning



Machine learning types





Although traditional ML models rely on supervised or unsupervised learning patterns, Generative AI models are typically pretrained through self-supervised learning.



Self-supervised learning

Unlabeled Training Data

The quick brown fox jumped over the lazy dog.



Remove Word
(create a pseudo label)

The quick brown fox jumped over the [] dog.



Model predicts word

The quick brown fox jumped over the [excited] dog.



Compare to pseudo label

excited $\neq$ lazy



Update model parameters and try again



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Domain 1: Fundamentals of AI and ML

Task Statement 1.2: Identify practical use cases for AI



Common Use Cases

Financial Fraud Detection

- Interpretability & Explainability crucial for regulations & fairness
- Traditional ML models preferred because of their transparency with clear decision rules

Healthcare Diagnostics

- Accuracy & Interpretability critical
- Traditional ML on structured data:
 - Patient records
 - Lab results
 - Imaging data
- Reliable diagnoses & treatment recommendations
- Decision process auditable by medical experts

Autonomous Vehicles

- Safety & Reliability paramount
- Traditional:
 - Computer Vision
 - Object Detection
 - Rule-based Systems
- Preferred over Generative AI
- Robust & Predictable in real-time environments

Manufacturing Quality Control

- Traditional ML on sensor data, images, structured data
- Detect defects, anomalies, quality issues
- Precise and consistent results crucial for:
 - Quality control
 - Process optimization

Generative AI models excel at tasks involving natural language processing, open-ended reasoning, and generation of human-like text or other modalities.

Common Use Cases

Credit Risk Assessment

- Transparency & Fairness critical for compliance
- Traditional models preferred:
 - Logistic Regression
 - Decision Trees
 - Credit Scoring
- Interpretable with clear decision criteria
- Avoid discrimination

Cybersecurity Threat Detection

- Traditional ML on network traffic, system logs, structured data
- Detect anomalies, intrusions, cyber threats
- High accuracy & reliability crucial:
 - Prevent false positives
 - Maintain system integrity

Predictive Maintenance

- Manufacturing, Aviation, Energy
- Sensor data, maintenance records
- Traditional models:
 - Time-series forecasting
 - Regression
- Accurate and interpretable
- Predict failures and maintenance needs

Traditional machine learning and AI approaches are preferred in domains where interpretability, transparency, robustness, and strict compliance with regulations are critical requirements.



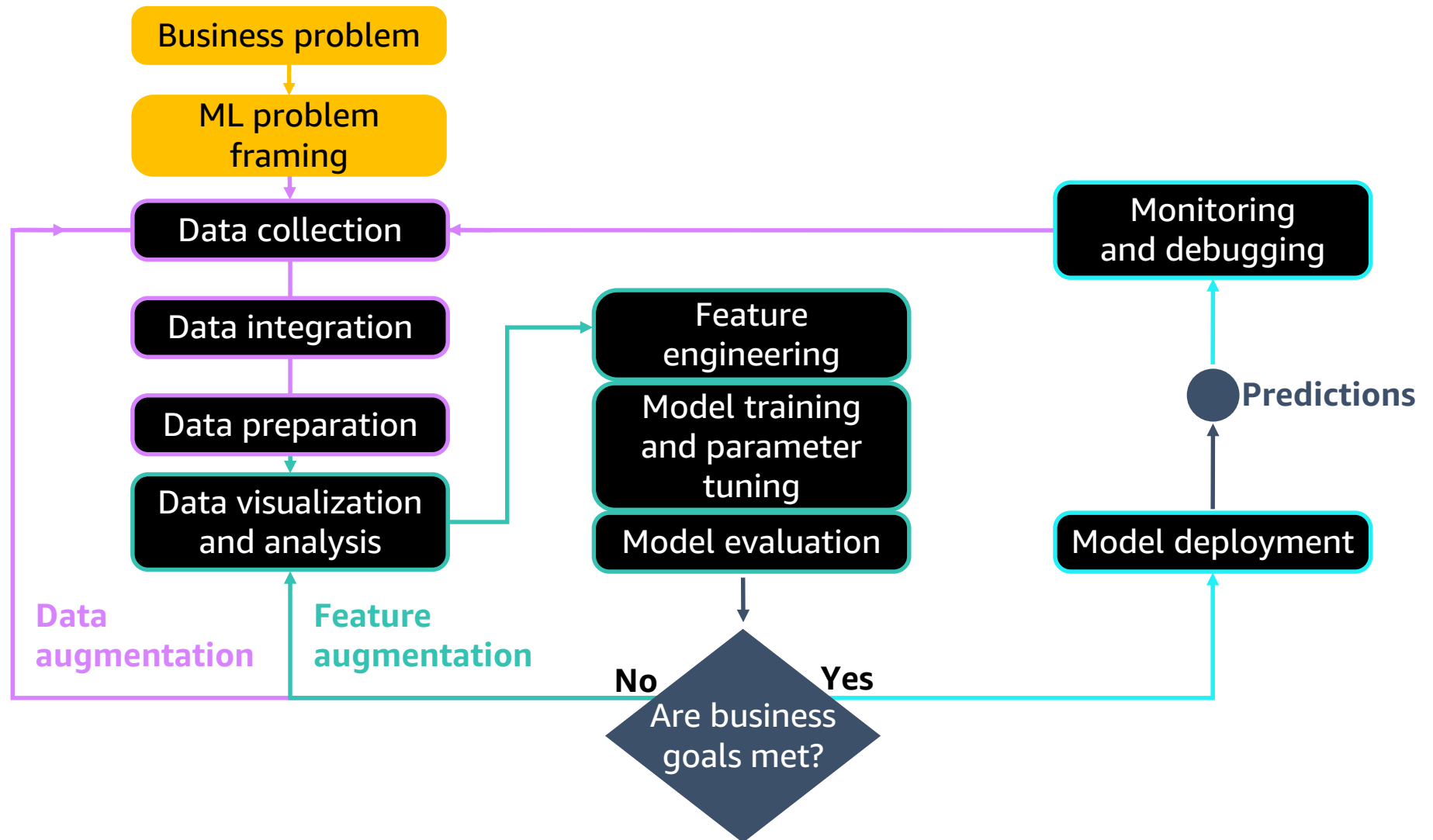
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Domain 1: Fundamentals of AI and ML

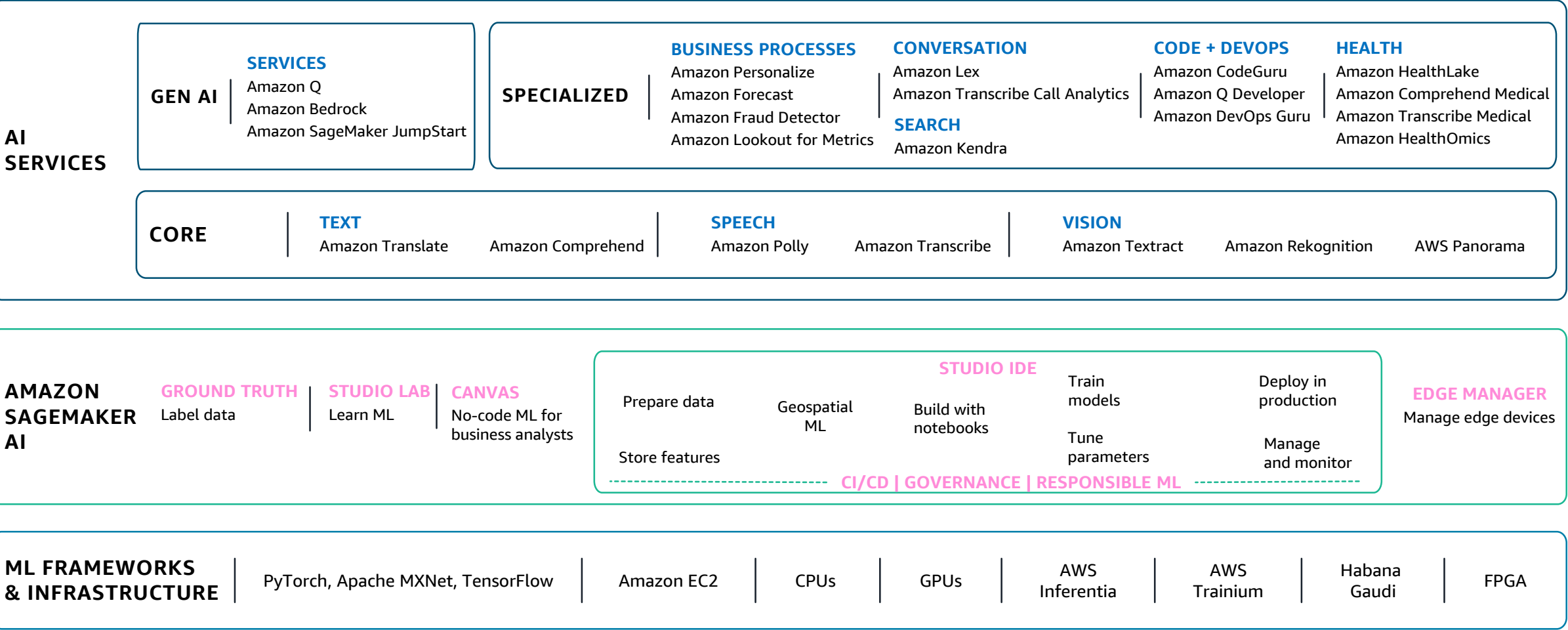
Task Statement 1.3: Describe the ML development lifecycle



ML process overview



Amazon AI/ML stack



Amazon SageMaker AI Overview

PREPARE

Geospatial

Visualize geospatial data

Ground Truth

Create high quality datasets for ML

Data Wrangler

Aggregate and prepare data for ML

Processing

Built-in Python, BYO R/Spark

Feature Store

Store, catalog, search, and reuse features

Clarify

Detect bias and understand model predictions

BUILD

Studio Notebooks & Notebook Instances

Fully managed Jupyter notebooks with elastic compute

Studio Lab

Free ML development environment

Built-in Algorithms

Integrated tabular, NLP, and vision algorithms

JumpStart

UI based discovery, training, and deployment of models, solutions, and examples

Autopilot

Automatically create ML models with full visibility

Bring Your Own

Bring your own container and algorithms

Local Mode

Test and prototype on your local machine

TRAIN & TUNE

Fully Managed Training

Broad hardware options, easy to setup and scale

Distributed Training Libraries

High performance training for large datasets and models

Training Compiler

Faster deep learning model training

Automatic Model Tuning

Hyperparameter optimization

Managed Spot Training

Reduce training cost by up to 90%

Debugger and Profiler

Debug and profile training runs

Experiments

Track, visualize, and share model artifacts across teams

Customization Support

Integrate with popular open source frameworks and libraries

DEPLOY & MANAGE

Fully Managed Deployment

Ultra low latency, high throughput inference

Real-Time Inference

For steady traffic patterns

Serverless Inference

For intermittent traffic patterns

Asynchronous Inference

For large payloads or long processing times

Batch Transform

For offline inference on batches of large datasets

Multi-Model Endpoints

Reduce cost by hosting multiple models per instance

Multi-Container Endpoints

Reduce cost by hosting multiple containers per instance

Shadow Testing

Validate model performance in production

Inference Recommender

Automatically select compute instance and configuration

Model Monitor

Maintain accuracy of deployed models

Kubernetes Operators & Components

Manage and monitor models on edge devices

Edge Manager

Manage and monitor models on edge devices

MLOps: Pipelines | Projects | Model Registry

Workflow automation, CI/CD for ML,
central model catalog

Canvas

Generate accurate machine learning
predictions—no code required

Studio | RStudio

Integrated development
environment (IDE) for ML

Governance

Model Cards | Dashboard |
Permissions

Amazon Rekognition



Amazon
Rekognition

Function

Automate image and video analysis with machine learning

Use cases

- Media analysis
- Identity verification
- Content moderation

Key Features

Labels	Custom labels
Content moderation	Text detection
Facial detection	Face search and verification
Celebrity recognition	Pathing

<https://aws.amazon.com/rekognition>

Highlighted
customer

C-SPAN

Amazon Textract



Amazon
Textract

Function

Extract any text and data from any document using machine learning and without manual effort

Use cases

- Create smart search indexes
- Build automated document processing workflows
- Maintain compliance in document archives

Benefits

Extract structured and unstructured data

Go beyond simple optical character recognition (OCR)

Security and compliance

Implement human reviews

Highlighted
customer

Filevine

<https://aws.amazon.com/textract/>

Amazon Comprehend



Amazon
Comprehend

Function

Discover insights and relationships in text

Use cases

- Call center analytics
- Index and search product reviews
- Personalize content on a website

Benefits

Get answers from text

Organize documents by topics

Train models on your own data

Support general and industry-specific text

<https://aws.amazon.com/comprehend>

Highlighted
customer

FINRA

Amazon Kendra

Amazon
Kendra

Function

Enterprise search service powered by machine learning

Use cases

- Improve access to internal knowledge
- Enhance sales and customer support services
- Help customers find information efficiently

Benefits

Ask natural-language questions, get immediate answers

Bring data together with a few clicks

Constantly improve search results

<https://aws.amazon.com/kendra>

Highlighted customers

[CORD-19 Search](#)

COVID-19 Open
Research Dataset
(CORD-19)

Amazon Personalize



Amazon
Personalize

Function

Create real-time, personalized user experiences fast, at scale

Use cases

- Retail – Help customers discover products
- Media and entertainment – Recommend new content, based on preference

Benefits

Deliver recommendations in real time

Implement personalized recommendations, in days

Personalize touchpoints along the customer journey

<https://aws.amazon.com/personalize>

Highlighted customers

Retail – Subway

Media and Entertainment – Coursera

Amazon Fraud Detector

Amazon
Fraud Detector

Function

Detect more online fraud faster

Use cases

Detect common types of fraud:

- New account
- Online payment
- Guest checkout
- Online service and loyalty program abuse

Benefits

Prevent and detect online fraud

Fraud detection in minutes

Customized for your unique business needs

<https://aws.amazon.com/fraud-detector>

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Highlighted
customers

Customers





Knowledge Check



Machine Learning Types

A company wants to develop a machine learning model to analyze customer feedback on their products. They have a dataset of 10,000 customer reviews, with each review labeled as either "positive" or "negative". The company will train the machine learning model to predict whether new customer reviews are “positive” or “negative”.

Which of the following machine learning types does this describe?

- A. Supervised Learning
- B. Unsupervised Learning
- C. Self-Supervised Learning
- D. Reinforcement Learning

Machine Learning Types

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Which of the following machine learning types does this describe?

A. Supervised Learning (Correct)

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C. Self-Supervised Learning

D. Reinforcement Learning

Recommendations

An e-commerce company needs to integrate tailored recommendations to customers based on their browsing and purchase history. The company needs to quickly implement this solution but their team has limited machine learning expertise.

Which of the following solutions would meet the requirements in the most operationally efficient way?

- A. Set up an EC2 cluster and train the model on the cluster. Use Amazon S3 to store your training data and model artifacts, and deploy the model on an EC2 instance behind an Elastic Load Balancer.
- B. Add the data to Amazon Personalize and then access real time recommendations via the personalization API.
- C. Use Amazon SageMaker AI to create a machine learning pipeline that includes data preparation, model training, and model deployment steps.
- D. Use Amazon Bedrock to generate personalized recommendations that would be shown to customers on the website.

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- B. Add the data to Amazon Personalize and then access real time recommendations via the personalization API. (Correct)**
- C. Use Amazon SageMaker AI to create a machine learning pipeline that includes data preparation, model training, and model deployment steps.
- D. Use Amazon Bedrock to generate personalized recommendations that would be shown to customers on the website.

ML Pipeline

A machine learning engineer wants to implement a machine learning pipeline on AWS to automate the process of training and deploying models. The pipeline should include data preprocessing, model training, and model deployment.

Which AWS service would the machine learning engineer use to orchestrate this machine learning pipeline?

- A. Amazon Rekognition
- B. Amazon Bedrock
- C. Amazon Q
- D. Amazon SageMaker AI

ML Pipeline

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Which AWS service would the machine learning engineer use to orchestrate this machine learning pipeline?

- A. Amazon Rekognition
- B. Amazon Bedrock
- C. Amazon Q
- D. Amazon SageMaker AI (Correct)**



Resource Sharing

Live walkthrough of useful documentation



Resources

<https://aws.amazon.com/what-is/artificial-intelligence/>

<https://aws.amazon.com/what-is/machine-learning/>

<https://aws.amazon.com/what-is/deep-learning/>

<https://aws.amazon.com/what-is/generative-ai/>

<https://aws.amazon.com/compare/the-difference-between-machine-learning-supervised-and-unsupervised/>

<https://aws.amazon.com/what-is/mlops/>

<https://aws.amazon.com/ai/machine-learning/>

<https://docs.aws.amazon.com/sagemaker/latest/dg/whatis.html>

<https://aws.amazon.com/sagemaker>

<https://aws.amazon.com/rekognition/>

<https://aws.amazon.com/personalize/>



Questions?

Thank you for attending this session

